

Metrizability Cannot Be Removed from the Closed-Form Criterion

A literature-implied negative answer to arXiv:1703.04883, Question 1

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Status. Full negative answer to the first loose end, obtained by applying a 1975 theorem of Dierolf. The metrizability hypothesis in Schmidt's Theorem 1.38 is essential: without it, assertions (i) and (ii) need not imply assertion (iii).

1 The source question

Schmidt's *Energy forms* [1] proves the following criterion as Theorem 1.38. Let (V, τ) be a complete topological vector space and let q be a quadratic form. If the form topology $(D(q), \tau_q)$ is metrizable, then any two of

- (i) q is closed;
- (ii) $\ker q$ is τ -closed and the canonical map

$$(D(q)/\ker q, q) \longrightarrow (V/\ker q, \tau/\ker q)$$

is continuous;

- (iii) $(D(q)/\ker q, q)$ is a Hilbert space

imply the third. Chapter 5, Section 5.1, Question 1 (printed page 163, arXiv PDF page 179) asks:

Does Theorem 1.38 hold without the assumption that τ_q is metrizable or are there counterexamples?

There are counterexamples.

2 The counterexample

Let

$$H_0 = c_{00} = \{(x_n)_{n \geq 1} : x_n = 0 \text{ for all but finitely many } n\}$$

with the norm inherited from ℓ^2 . This is a Hausdorff locally convex space but is not complete.

Dierolf's Theorem 4 [2, p. 76] says that every topological vector space is a quotient of a complete Hausdorff topological vector space. Hence there are a complete Hausdorff topological vector space (Z, τ) and a quotient map

$$\pi : Z \longrightarrow H_0.$$

Write $L = \ker \pi$. Since $H_0 \simeq Z/L$ is Hausdorff, L is τ -closed. Define the finite quadratic form

$$q : Z \longrightarrow [0, \infty), \quad q(z) = \|\pi z\|_2^2.$$

Theorem 1. *For this q , assertions (i) and (ii) of Schmidt's Theorem 1.38 hold, while assertion (iii) fails.*

Proof. We have $D(q) = Z$ and $\ker q = L$. The seminorm $q^{1/2} = \|\pi(\cdot)\|_2$ is τ -continuous, because the quotient identification $Z/L \simeq H_0$ is a homeomorphism. The form topology is the smallest vector topology containing both τ and the topology induced by $q^{1/2}$; consequently

$$\tau_q = \tau.$$

Since Z is complete, $(D(q), \tau_q)$ is complete. Thus q is closed, proving (i).

We already observed that L is closed. Under the quotient identification, both

$$(D(q)/L, q) \quad \text{and} \quad (Z/L, \tau/L)$$

are precisely H_0 with its ℓ^2 norm topology. The canonical map in (ii) is therefore a topological isomorphism (indeed, an isometry on the form-norm side). This proves (ii).

Finally, $(D(q)/L, q)$ is the incomplete pre-Hilbert space H_0 . For example, the finite truncations of $(1/n)_{n \geq 1} \in \ell^2 \setminus c_{00}$ form a Cauchy sequence in H_0 without a limit there. Hence (iii) fails. $\square$

The example genuinely lies outside the omitted hypothesis. If $\tau_q = \tau$ were metrizable, then the quotient of the complete metrizable topological vector space Z by its closed subspace L would be complete, contradicting $Z/L \simeq H_0$.

3 Provenance and scope

This is a literature-implied answer: Dierolf's theorem predates the 2017 question by four decades but does not mention quadratic forms. The pullback construction above is the direct identification that closes the question. A bounded search through 13 August 2026 found the source question, Dierolf's theorem, and standard discussions of incomplete quotients of complete nonmetrizable spaces, but no explicit publication of this quadratic-form application. No priority claim is made.

The argument answers only Question 1 in Schmidt's loose-end chapter. It does not settle the later questions about Hilbert quotients of energy forms or local Poincaré inequalities.

References

- [1] M. Schmidt, *Energy forms*, Ph.D. thesis, Friedrich Schiller University Jena, 2017; arXiv:1703.04883v1. See Theorem 1.38 and Chapter 5, Question 1.
- [2] S. Dierolf, *Über Quotienten vollständiger topologischer Vektorräume*, Manuscripta Math. **17** (1975), 73–78. See Theorem 4, p. 76.